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Regular End Semester Examination - Summer 2022

Course: B. Tech. Branch: Chemical Semester: VIII

Subject: Process Control: Design, Analysis and assessment (BTCHC 801A)

Max Marks: 60 Date: 04/072022

Instructions to the Students:

1. *All the questions are compulsory.*
2. *Assume suitable data wherever necessary and mention it clearly.*

Q. 1 **A** Differentiate between feedback and feed forward control strategies. **(6 M)**
Develop a feedback control and feed forward control systems for controlling the temperature of liquid inside the stirred tank, which is heated by steam passing through the steam coil immersed inside the tank. Identify the different variables involved in the process.

B Assuming that the thermometer is a first order system, thermometer **(6 M)**
showing steady state temperature of 30°C is suddenly immersed into a hot bath at 150°C. If the time constant of thermometer is 7 sec, determine the following:

i. Thermometer reading after 7 sec.

ii. Time required to read 90°C on Thermometer

Q. 2 Solve any *TWO.*

A Develop a transfer function for Mercury Thermometer & mention the **(6 M)**
assumptions.

B Discuss the incentives of process modelling and control along with that **(6 M)**
explain SISO systems with examples from chemical engineering field.

C What are the various forcing functions used to study the dynamic **(6 M)**
response of a system? Explain each with neat sketch.

Q. 3 Explain the root locus and its significance. Draw the root locus of a **(12 M)**
closed-loop system with the following characteristics

$$G_p(s) = \frac{1}{(2s+1)(s+1)}, \; G_c(s) = Kc, \; G_m(s) = G_f(s) = 1.$$

Indicate which segment of the root locus (i.e. value of K_c) yield
(a) over damped, (b) underdamped and (c) critically damped closed loop response.

OR

Q. 3 Define the controller tuning and discuss the following tuning techniques **(12 M)**

i. Time integral performance criteria

ii. Process Reaction Curve method

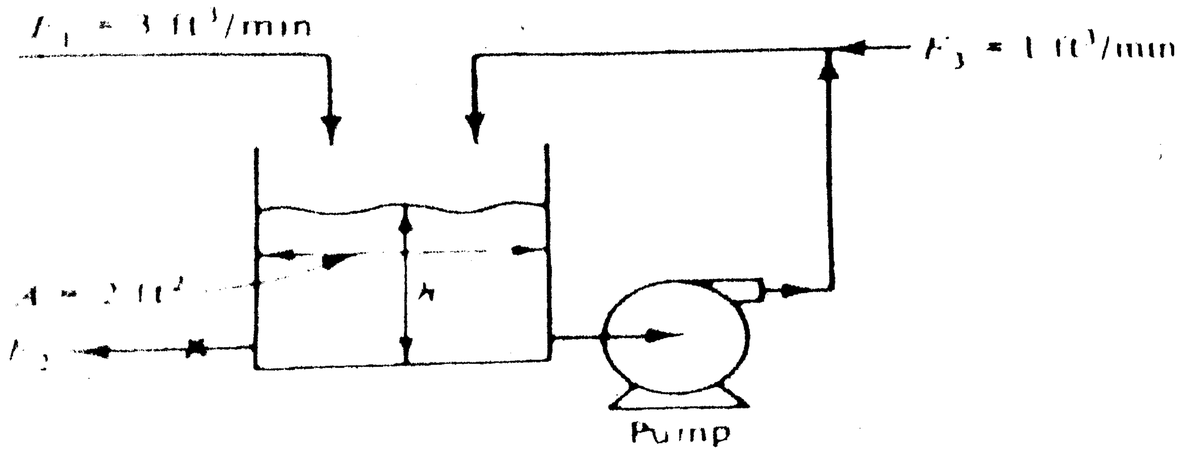
iii. Zeigler Nichols Method

Q. 4 Solve any *TWO*.

A Explain the term offset and why proportional controller will always show an offset for regulatory problems? And also prove with example that integral mode will always remove offset from a servo problem. **(6 M)**

B Derive the expression for Amplitude Ratio and phase angle for pure capacitive system and also plot asymptotic Bode diagram for the same. **(6 M)**

C Consider a storage tank as shown in Figure1. Develop a transfer function between liquid level and inlet stream. Determine the time constant and process gain. Assume that the exit flow rates are linear function of the liquid level. i.e. F= 1.5 (ft^3/ft. min) × h (ft). **(6 M)**



Q. 5 Derive the transfer function for both Non-interacting and interacting tanks and explain how the middle term of denominator polynomial of both the transfer function is different. **(12 M)**